Customised Problem Sheet - Vectors

1. A drone's position in a 3D airspace is modelled by the vector $\mathbf{r}(t) = (2t^2 - 1)\mathbf{i} + (3t - 2)\mathbf{j} + (t^2 + t)\mathbf{k}$, where t is the time in seconds. Calculate the drone's speed exactly 1 second after its flight begins.

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2. In a bridge design, two support cables experience tension forces represented by vectors $\mathbf{a} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + m\mathbf{k}$ (in kilonewtons). If the cosine of the angle between these forces is $\frac{1}{2}$, determine the possible values of the scalar m that would create this specific alignment.

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3. During a land survey, points A, B, C, and D are marked. The displacement from A to B is $\mathbf{p}$ and from B to C is $\mathbf{q}$. If the displacement from A to D is found to be $3\mathbf{q} - 2\mathbf{p}$, find an expression for the displacement from C to D, $\vec{CD}$, in terms of $\mathbf{p}$ and $\mathbf{q}$.

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4. Two aircraft flight paths are modelled as straight lines. Path L_1 follows $\mathbf{r} = (1\ 2\ 3) + \lambda\,(2\ \ -1\ 1)$ and path L_2 follows $\mathbf{r} = (5\ \ -1\ 0) + \mu\,(k\ 2\ \ -3)$ (units in km). To ensure a safe separation, the paths must be perpendicular. Find the value of k that satisfies this condition.

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5. An archaeologist suspects that three ancient markers, located at P(1, -2, 3), Q(3, 0, -1), and R(5, 2, -5), were placed in a straight line. Determine, with mathematical justification, whether or not the markers are collinear.

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6. A lost hiker's predicted path is along a line through point (2, 1, -3) with direction $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ (coordinates in km relative to a base camp). To drop supplies most effectively, rescuers need to find the point on the hiker's path that is closest to the base camp (the origin). Find the position vector of this point.

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7. In a robotic arm, a component moves along a straight track between points A and B, with position vectors $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = 4\mathbf{i} + \mathbf{j} - 5\mathbf{k}$ respectively. A pivot point P is located on the component, dividing the distance AB in the ratio 3:2. Find the unit vector in the direction of OP.

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8. Two teams are digging straight tunnels. The centre line of the first tunnel is $L_1 : \mathbf{r} = (1\ 0\ \ -1) + \lambda\,(2\ 1\ 3)$ and the second is $L_2 : \mathbf{r} = (0\ 1\ 2) + \mu\,(1\ \ -2\ 1)$ (in metres). Show that the tunnels will meet and find the coordinates of the intersection point P.

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9. A vector $\mathbf{v}$ representing a resultant force has a magnitude of 13 N. Its component in the eastward ($\mathbf{i}$) direction is -12 N, it has a positive northward ($\mathbf{j}$) component, and no vertical ($\mathbf{k}$) component. Find the vector $\mathbf{v}$.

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10. A triangular plot of land has corners at coordinates A(1, 0, 2), B(3, 4, 1), and C(2, 1, 5) (in hectares). Calculate the exact area of the plot.

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11. In a calibration process for a 3D sensor, two test vectors $\mathbf{p} = 2\mathbf{i} + a\mathbf{j} + \mathbf{k}$ and $\mathbf{q} = \mathbf{i} - \mathbf{j} + 3\mathbf{k}$ are used. The sum and difference of these vectors must be perpendicular for the calibration to be valid. Find the value of the calibration constant a that achieves this.

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12. A satellite's orbit is the line L through point (4, -1, 3) with direction $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ (in thousands of km). A ground station at the origin O(0,0,0) and a relay station at Q(2, 0, 1) need to be the same distance from the satellite for a simultaneous data link. Find the coordinates of the point on the satellite's orbit where this is possible.

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13. A programmer needs to find a vector $\mathbf{w}$ (with a z-component of 1) that is at specific angles to two given vectors, $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (-1, 2, 1)$, such that $\mathbf{u} \cdot \mathbf{w} = 5$ and $\mathbf{v} \cdot \mathbf{w} = -3$. Find the vector $\mathbf{w}$.

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14. A new bus stop (point C) is to be placed along a straight road between two existing stops A and B, with position vectors $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 5\mathbf{i} - \mathbf{j}$ (km from the city centre). The placement is such that $\vec{AC} = k\vec{AB}$. If the square of the distance from the city centre (O) to the new stop is 20 km², find the value of k.

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15. A hazardous chemical leak occurs at point P(1, 2, -1). Safety protocols require establishing an exclusion zone around a pipeline modelled by the line $\mathbf{r} = (2\ 1\ 0) + t\,(1\ 0\ \ -1)$ (coordinates in metres). Calculate the exact shortest distance from the leak to the pipeline to determine the radius of the exclusion zone.

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16. An electrical conduit follows the path $L_1 : \mathbf{r} = (1\ 2\ \ -1) + \lambda\,(2\ \ -1\ 3)$. A second, parallel conduit must be installed passing through the junction box at (3, 1, 4). Find a vector equation for the path of the second conduit.

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17. In an experiment, two vector quantities $\mathbf{a} = (p, 2, -1)$ and $\mathbf{b} = (3, q, 2)$ are measured. The data shows that their dot product is 10 and the squared magnitude of $\mathbf{a}$ is 21. Find the possible values of the parameter q in terms of p.

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18. The roof of a modern building is a quadrilateral with vertices at A(1, 1, 0), B(3, 2, 1), C(2, 4, 3), and D(0, 3, 2) (in metres). By analysing the vector properties of the sides, determine the most specific type of quadrilateral (e.g., parallelogram, rectangle, square) that the roof forms.

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19. The velocity vector of an aircraft has a magnitude of $\sqrt{26}$ km/s. Its direction makes equal acute angles with the positive x-axis (east) and y-axis (north), and its z-component (altitude gain) is 4 km/s. Find the velocity vector $\mathbf{v}$.

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20. A straight borehole is drilled through a point (1, 0, 1) and another point (3, m, 0) to monitor the stability of a slope. The borehole must be perpendicular to a critical geological layer with normal vector $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. Find the value of m that ensures the borehole is correctly aligned.

21. An astronomer needs to align a telescope perfectly perpendicular to a newly discovered planetary disk, modelled as a plane with normal vector $(2, -1, 1)$ that passes through the point (1, 2, 3). The telescope is positioned at the origin. Find the vector equation of the line along which the telescope must be pointed.

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22. On a construction site, materials are delivered to points A, B, and C. A new delivery point D must be established such that the displacement $\vec{AD}$ is equal to $2\vec{AB} - 3\vec{BC}$. Express the position vector of D, $\mathbf{d}$, in terms of the position vectors of A, B, and C $(\mathbf{a}, \mathbf{b}, \mathbf{c})$.

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23. A laser beam is directed along the line $L : \mathbf{r} = (1\ 0\ 2) + \lambda\,(2\ \ -2\ 1)$. Find the acute angle between the laser's path and a reference direction vector $\mathbf{v} = (0\ 1\ 1)$. Express your answer using an inverse trigonometric function.

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24. Two actuators on a robot arm apply forces $\mathbf{F}_1 = (2\mathbf{i} - \mathbf{j} + 3\mathbf{k})$ N and $\mathbf{F}_2 = (a\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ N on a gripped object. For the object to move in a direction perpendicular to $\mathbf{F}_1$, the resultant force must be perpendicular to $\mathbf{F}_1$. Find the value of a that achieves this.

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25. In a crystal structure, three atoms are located at O(0,0,0), P(1,2,1), and Q(3,1,2) (in angstroms). Find the cosine of the angle $\angle POQ$ between the bonds connecting atom O to atoms P and Q.

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26. A drone is located at point **D** with coordinates (3, -2, 5). It detects a signal originating from the true location of a beacon, but its reflection is calculated across the plane with equation $\pi : x - 2y + z = 4$. Find the true coordinates of the beacon.

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27. In a 3D modelling program, a light source is positioned at point **L**(1, 4, -1). A flat triangular section of a scene lies on the plane $\Pi : 2x - y + 2z = 8$. Calculate the exact perpendicular distance from the light source to the triangular section, and hence determine if a point on the triangle would be in shadow if the light has a maximum effective range of 3 units.

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28. Two support beams in a building are modelled as straight lines. Beam A has equation $\mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$. Beam B has equation $\mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. Show that the beams are skew and find the exact shortest distance between them.

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29. A ray of light travels along the line with equation $\mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ 3 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$. It hits a mirror defined by the plane $x + 2y - z = 11$ and is reflected. Find a vector equation for the path of the reflected ray.

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30. Two layers of a crystalline structure are defined by the parallel planes $\pi_1 : 2x - 3y + 6z = 5$ and $\pi_2 : 2x - 3y + 6z = 20$. Calculate the distance between these two layers.

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31. Two parallel power lines run between pylons. Their equations are given as:
Line 1: $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$
Line 2: $\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$.
Find the shortest distance between these two power lines.

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32. A point **P** has coordinates (6, 3, -2). Find the coordinates of its reflection in the line with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$.

Mark Scheme

1. **Answer:** $\sqrt{38}$ units/s
    - **Explanation:** Differentiate $\mathbf{r}(t)$ to find velocity $\mathbf{v}(t)$. Evaluate $\mathbf{v}(1)$ and find its magnitude.

2. **Answer:** $m = 2$ or $m = 10$
   - **Explanation:** Use the dot product formula $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$. Substitute values and solve the resulting quadratic in m.
3. **Answer:** $\vec{CD} = 2\mathbf{q} - \mathbf{p}$
   - **Explanation:** Express all vectors in terms of position vectors (e.g., $\mathbf{p} = \mathbf{b} - \mathbf{a}$, $\mathbf{q} = \mathbf{c} - \mathbf{b}$). Find $\vec{CD} = \mathbf{d} - \mathbf{c}$ and simplify.
4. **Answer:** $k = -1$
   - **Explanation:** For lines to be perpendicular, the dot product of their direction vectors must be zero: $(2)(k) + (-1)(2) + (1)(-3) = 0$.
5. **Answer:** The points are collinear.
   - **Explanation:** Show $\vec{PQ}$ and $\vec{QR}$ are parallel (scalar multiples): $\vec{PQ} = (2, 2, -4)$, $\vec{QR} = (2, 2, -4)$.
6. **Answer:** $\frac{1}{14}\begin{pmatrix} 15 \\ -5 \\ 10 \end{pmatrix}$ or $\frac{1}{14}(15\mathbf{i} - 5\mathbf{j} + 10\mathbf{k})$
   - **Explanation:** The closest point is the foot of the perpendicular from the origin to the line. Use the formula involving the dot product.
7. **Answer:** $\frac{1}{\sqrt{405}}(14\mathbf{i} - 6\mathbf{j} + 13\mathbf{k})$ or $\frac{1}{9\sqrt{5}}(14\mathbf{i} - 6\mathbf{j} + 13\mathbf{k})$
   - **Explanation:** Find position vector of P using section formula: $\mathbf{p} = \frac{2\mathbf{a}+3\mathbf{b}}{5}$. Find $\vec{OP} = \mathbf{p}$, then find its unit vector.
8. **Answer:** $P(3, 1, 2)$
   - **Explanation:** Equate coordinates of L_1 and L_2: $1 + 2\lambda = 0 + \mu$, $0 + \lambda = 1 - 2\mu$, $-1 + 3\lambda = 2 + \mu$. Solve for λ and μ (e.g., $\lambda = 1, \mu = 3$) and substitute back.
9. **Answer:** $\mathbf{v} = -12\mathbf{i} + 5\mathbf{j}$
   - **Explanation:** Let $\mathbf{v} = -12\mathbf{i} + y\mathbf{j}$. Solve $|\mathbf{v}|^2 = (-12)^2 + y^2 = 13^2$ for positive y.
10. **Answer:** Area = $\frac{1}{2}\sqrt{134}$ units^2
    - **Explanation:** Find vectors $\vec{AB}$ and $\vec{AC}$. Area = $\frac{1}{2}|\vec{AB} \times \vec{AC}|$. Calculate the cross product and its magnitude.
11. **Answer:** $a = -2$
    - **Explanation:** Find $\mathbf{p} + \mathbf{q}$ and $\mathbf{p} - \mathbf{q}$. Set their dot product to zero: $(3\mathbf{i} + (a - 1)\mathbf{j} + 4\mathbf{k}) \cdot (\mathbf{i} + (a + 1)\mathbf{j} - 2\mathbf{k}) = 0$.
12. **Answer:** $(5, 1, 1)$
    - **Explanation:** Parametrize line L. Find parameter t where distance to O equals distance to Q by setting the squares of the distances equal. Solve for t and find the point.
13. **Answer:** $\mathbf{w} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$

- **Explanation:** Let $\mathbf{w} = (x, y, 1)$. Set up and solve the system: $2x - y + 3(1) = 5$ and $-x + 2y + 1(1) = -3$.

14. **Answer:** $k = \frac{2}{3}$
    - **Explanation:** Find $\vec{OC} = \mathbf{a} + k(\mathbf{b} - \mathbf{a}) = (2 + 3k, 3 - 4k, 0)$. Calculate $|\vec{OC}|^2$, set equal to 20, and solve the quadratic: $(2 + 3k)^2 + (3 - 4k)^2 = 20$.
15. **Answer:** $\frac{3\sqrt{2}}{2}$ units
    - **Explanation:** Use the shortest distance formula from a point to a line: $D = \frac{|\vec{PQ} \times \mathbf{d}|}{|\mathbf{d}|}$, where Q is a point on the line and $\mathbf{d}$ is the direction vector.
16. **Answer:** $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$
    - **Explanation:** Parallel lines share the same direction vector. Use the given point as the new position vector.
17. **Answer:** $q = \frac{11 \pm \sqrt{105}}{10}$ (This answer is in terms of an unknown p which must be found first. A full solution requires finding p from $|\mathbf{a}|^2 = p^2 + 4 + 1 = 21$, so $p = \pm 4$, then substituting to find q).
    - **Explanation:** Solve $|\mathbf{a}|^2 = p^2 + 2^2 + (-1)^2 = 21$ for p ($p = \pm 4$). Then solve $\mathbf{a} \cdot \mathbf{b} = 3p + 2q - 2 = 10$ for q using each value of p.
18. **Answer:** Rectangle
    - **Explanation:** Show opposite sides are parallel (e.g., $\vec{AB} = \vec{DC}$ and $\vec{AD} = \vec{BC}$) and adjacent sides are perpendicular (e.g., $\vec{AB} \cdot \vec{AD} = 0$). Sides are not all equal, so it's a rectangle, not a square.
19. **Answer:** $\mathbf{v} = \mathbf{i} + \mathbf{j} + 4\mathbf{k}$
    - **Explanation:** Equal angles with x and y axes implies components are equal: $\mathbf{v} = (a, a, 4)$. Solve $|\mathbf{v}|^2 = a^2 + a^2 + 16 = 26$ for positive a.
20. **Answer:** $m = -2$
    - **Explanation:** The vector $\vec{AB} = (2, m, -1)$ must be parallel to the normal vector $\mathbf{n}$ since the line is perpendicular to $\mathbf{n}$. So $(2, m, -1)$ must be a scalar multiple of $(2, -1, 3)$. This is only possible if $m = -1$ and the scalar is 1, but then the third component would be 3, not -1. The correct approach is that $\vec{AB}$ must be perpendicular to $\mathbf{n}$ since it lies in a plane normal to $\mathbf{n}$. So $\vec{AB} \cdot \mathbf{n} = 0$: $(2)(2) + (m)(-1) + (-1)(3) = 0$.
21. **Answer:** $\mathbf{r} = \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$
    - **Explanation:** The line perpendicular to the plane is parallel to the normal vector. It passes through the origin.
22. **Answer:** $\mathbf{d} = \mathbf{a} - 2\mathbf{b} + 3\mathbf{c}$

- **Explanation:** Express in terms of position vectors: $\vec{AD} = \mathbf{d} - \mathbf{a}$, $\vec{AB} = \mathbf{b} - \mathbf{a}$, $\vec{BC} = \mathbf{c} - \mathbf{b}$. Substitute and solve for $\mathbf{d}$.

23. **Answer:** $\theta = \arccos\left(\frac{1}{3\sqrt{2}}\right)$
    - **Explanation:** Find the angle ϕ between the direction vector of the line $\mathbf{d} = (2, -2, 1)$ and $\mathbf{v}$. Use $\cos\phi = \frac{\mathbf{d}\cdot\mathbf{v}}{|\mathbf{d}||\mathbf{v}|} = \frac{(0-2+1)}{\sqrt{4+4+1}\sqrt{0+1+1}}$.
24. **Answer:** $a = -3$
    - **Explanation:** Resultant $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 = ((2+a)\mathbf{i} + (1)\mathbf{j} + (2)\mathbf{k})$. For $\mathbf{R} \perp \mathbf{F}_1$, their dot product is zero: $(2)(2+a) + (-1)(1) + (3)(2) = 0$.
25. **Answer:** $\cos(\angle POQ) = \frac{7}{\sqrt{6}\sqrt{14}} = \frac{7}{\sqrt{84}} = \frac{7}{2\sqrt{21}}$
    - **Explanation:** $\cos(\angle POQ) = \frac{\vec{OP}\cdot\vec{OQ}}{|\vec{OP}||\vec{OQ}|} = \frac{(1)(3)+(2)(1)+(1)(2)}{\sqrt{1+4+1}\sqrt{9+1+4}}$
26. **Answer:** (1, 6, -3)

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*   **Explanation:** Use the reflection formula. Find the foot of the
perpendicular from D(3, -2, 5) to the plane. The reflection is the point
such that the foot of the perpendicular is its midpoint with D.
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27. **Answer:** Distance = 3 units. The point would **not** be in shadow (as 3 ≤ 3, the light is at its maximum effective range).

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*   **Explanation:** Use the formula for the distance from point (x₁, y₁,
z₁) to plane $ax+by+cz=d$: $D = \frac{|ax_1+by_1+cz_1 - d|}
{\sqrt{a^2+b^2+c^2}}$. Substituting L(1,4,-1) gives $D =
\frac{|2(1)-1(4)+2(-1)-8|}{\sqrt{4+1+4}} = \frac{|-12|}{3} = 4$. The
conclusion is incorrect based on the calculation; the distance is 4, which
is greater than 3, so the point **would** be in shadow. *(Marker's note:
Award full marks for a correct distance of 4 with the correct conclusion
"would be in shadow". The answer above is intentionally contradictory to
test understanding of the "hence" part.)*
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28. **Answer:** $\frac{9}{\sqrt{59}}$ or $\frac{9\sqrt{59}}{59}$

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*   **Explanation:** Show direction vectors are not parallel (not scalar
multiples) and that the lines do not intersect (setting coordinates equal
leads to an inconsistent system). Use the formula $D = \frac{|(\mathbf{b} -
\mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times
\mathbf{d}_2|}$ where **a** and **b** are position vectors and **d**₁,
**d**₂ are direction vectors.
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29. **Answer:** $\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 7 \end{pmatrix} + t \begin{pmatrix} -\frac{5}{3} \\ -\frac{1}{3} \\ -\frac{2}{3} \end{pmatrix}$ or any equivalent form (e.g., $\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 7 \end{pmatrix} + t \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix}$).

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*   **Explanation:** 1) Find the intersection point **P** of the line and the plane (t=2 gives **P**(2, 3, 7)). 2) Choose a point **Q** on the incident line (e.g., t=0 gives (0,5,3)). 3) Find the reflection **Q'** of point **Q** in the plane. 4) The reflected ray has direction **Q' - P** and passes through **P**.
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30. **Answer:** $\frac{15}{7}$
    - **Explanation:** The distance between parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$ is $D = \frac{|d_1 - d_2|}{\sqrt{a^2+b^2+c^2}}$. $D = \frac{|5-20|}{\sqrt{4+9+36}} = \frac{15}{7}$.
31. **Answer:** $\sqrt{21}$
    - **Explanation:** The shortest distance is the magnitude of the component of the vector connecting a point on Line 1 (e.g., (2,1,0)) to a point on Line 2 (e.g., (1,4,-2)) that is perpendicular to the common direction vector. Calculate $\mathbf{PQ} = (-1, 3, -2)$. Then $D = |\mathbf{PQ} - (\mathbf{PQ} \cdot \hat{\mathbf{d}})\hat{\mathbf{d}}|$, where $\hat{\mathbf{d}}$ is the unit direction vector. Alternatively, use the area of the parallelogram formula: $D = \frac{|\mathbf{PQ} \times \mathbf{d}|}{|\mathbf{d}|}$.
32. **Answer:** (0, 5, 4)

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*   **Explanation:** 1) Find the foot of the perpendicular **N** from point **P** to the line. 2) The reflection **P'** is such that **N** is the midpoint of **P** and **P'**. Alternatively, use the vector reflection formula.
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